

Table T3. Single-feature predictive power: held-out accuracy and mutual information for the fifteen highest-ranked features of each benchmark.

dataset	feature	accuracy	precision	recall	f1	mutual_information
PhiUSIIL	URLSimilarityIndex	0.996737	1	0.992344	0.996157	0.662216
PhiUSIIL	NoOfExternalRef	0.961134	0.939419	0.97144	0.955161	0.565449
PhiUSIIL	LineOfCode	0.960935	0.965348	0.942148	0.953607	0.595168
PhiUSIIL	NoOfSelfRef	0.948708	0.901977	0.986885	0.942523	0.529394
PhiUSIIL	NoOfImage	0.942126	0.942343	0.920511	0.931299	0.549451
PhiUSIIL	NoOfJS	0.93197	0.930261	0.908461	0.919232	0.504826
PhiUSIIL	NoOfCSS	0.90187	0.96787	0.796152	0.873653	0.455661
PhiUSIIL	HasSocialNet	0.882663	0.786372	0.99494	0.878446	0.391132
PhiUSIIL	HasCopyrightInfo	0.867429	0.788374	0.941682	0.858235	0.32197
PhiUSIIL	LargestLineLength	0.838464	0.955731	0.651088	0.774531	0.458574
PhiUSIIL	HasDescription	0.832846	0.732681	0.95686	0.829898	0.280625
PhiUSIIL	NoOf0therSpecialCharsInURL	0.793838	0.935227	0.554624	0.69631	0.234781
PhiUSIIL	IsHTTPS	0.791569	1	0.510885	0.676272	0.234807
PhiUSIIL	DomainTitleMatchScore	0.789866	0.719121	0.831769	0.771354	0.191269
PhiUSIIL	IsResponsive	0.781412	0.777037	0.683044	0.727015	0.16657
UCI Phishing Websites	sslfinal_state	0.873576	0.905213	0.843267	0.873143	0.344777
UCI Phishing Websites	url_of_anchor	0.81549	0.996587	0.644592	0.782842	0.310446
UCI Phishing Websites	having_sub_domain	0.68451	0.650685	0.838852	0.732883	0.0806268
UCI Phishing Websites	web_traffic	0.667426	0.670913	0.697572	0.683983	0.0585225
UCI Phishing Websites	prefix_suffix	0.637813	0.587549	1	0.740196	0.104922
UCI Phishing Websites	sfh	0.615034	0.588872	0.84106	0.692727	0.0462649
UCI Phishing Websites	links_in_tags	0.596811	0.646018	0.483444	0.55303	0.0393409
UCI Phishing Websites	request_url	0.591116	0.623684	0.523179	0.569028	0.0326477
UCI Phishing Websites	domain_registration_length	0.584282	0.650685	0.419426	0.510067	0.0179795
UCI Phishing Websites	having_ip_address	0.571754	0.595533	0.529801	0.560748	0
UCI Phishing Websites	links_pointing_to_page	0.563781	0.57085	0.622517	0.595565	0.00555532
UCI Phishing Websites	google_index	0.531891	0.63125	0.222958	0.329527	0
UCI Phishing Websites	age_of_domain	0.529613	0.544444	0.540839	0.542636	0.00909955
UCI Phishing Websites	abnormal_url	0.529613	0.526882	0.865342	0.654971	0.0129949
UCI Phishing Websites	statistical_report	0.525057	0.636364	0.18543	0.287179	0

Table T3. Single-feature predictive power: held-out accuracy and mutual information for the fifteen highest-ranked features of each benchmark. (continued)

Mendeley (Hannousse)	google_index	0.85481	0.861127	0.299412
Mendeley (Hannousse)	web_traffic	0.787755	0.786385	0.298463
Mendeley (Hannousse)	nb_hyperlinks	0.777259	0.79831	0.233622
Mendeley (Hannousse)	page_rank	0.771429	0.761557	0.221179
Mendeley (Hannousse)	ratio_intHyperlinks	0.728863	0.697856	0.244491
Mendeley (Hannousse)	nb_www	0.727697	0.7427	0.110777
Mendeley (Hannousse)	ratio_extHyperlinks	0.724781	0.711491	0.240274
Mendeley (Hannousse)	safe_anchor	0.720117	0.73064	0.17924
Mendeley (Hannousse)	longest_word_path	0.697376	0.667094	0.119986
Mendeley (Hannousse)	links_in_tags	0.690379	0.70936	0.120908
Mendeley (Hannousse)	ratio_extRedirection	0.689213	0.700393	0.120895
Mendeley (Hannousse)	domain_age	0.688047	0.655505	0.279648
Mendeley (Hannousse)	shortest_word_host	0.68105	0.680677	0.0738478
Mendeley (Hannousse)	char_repeat	0.674052	0.680023	0.0881938
Mendeley (Hannousse)	ratio_digits_url	0.673469	0.653465	0.102036
Mendeley (Tan)	PctExtHyperlinks	0.888667	0.887693	0.469465
Mendeley (Tan)	PctExtResourceUrls	0.779333	0.753537	0.293879
Mendeley (Tan)	PctExtNullSelfRedirectHyperlink sRT	0.779333	0.725766	0.216606
Mendeley (Tan)	PctNullSelfRedirectHyperlinks	0.735333	0.780055	0.234187
Mendeley (Tan)	ExtMetaScriptLinkRT	0.730667	0.754854	0.121785
Mendeley (Tan)	NumNumericChars	0.722667	0.740971	0.124515
Mendeley (Tan)	FrequentDomainNameMismatch	0.698667	0.583026	0.131842
Mendeley (Tan)	NumDash	0.669333	0.725967	0.11325
Mendeley (Tan)	SubmitInfoToEmail	0.634667	0.729516	0.0772315
Mendeley (Tan)	InsecureForms	0.624667	0.718922	0.0586642
Mendeley (Tan)	PathLevel	0.623333	0.668622	0.062891
Mendeley (Tan)	IframeOrFrame	0.62	0.675029	0.0355051
Mendeley (Tan)	UrlLength	0.617333	0.655048	0.0419254
Mendeley (Tan)	NumDots	0.615333	0.561217	0.0611993
Mendeley (Tan)	PathLength	0.604	0.692547	0.0546985

Table T3. Single-feature predictive power: held-out accuracy and mutual information for the fifteen highest-ranked features of each benchmark. (continued)

UCI Website Phishing	SFH	0.832512	0.858333	0.256211
UCI Website Phishing	SSLfinal_State	0.788177	0.813853	0.167353
UCI Website Phishing	popUpWindow	0.743842	0.788618	0.146957
UCI Website Phishing	URL_Length	0.615764	0.704545	0.0175735
UCI Website Phishing	web_traffic	0.605911	0.692308	0.0371276
UCI Website Phishing	having_IP_Address	0.596059	0.746914	0
UCI Website Phishing	Request_URL	0.591133	0.655602	0.0467234
UCI Website Phishing	age_of_domain	0.586207	0.658537	0
UCI Website Phishing	URL_of_Anchor	0.576355	0.62931	0.0406747
UCI Website Phishing (variant)	SFH	0.87766	0.895928	0.347106
UCI Website Phishing (variant)	SSLfinal_State	0.835106	0.85446	0.158212
UCI Website Phishing (variant)	popUpWindow	0.75	0.78733	0.173193
UCI Website Phishing (variant)	URL_of_Anchor	0.62766	0.678899	0.0638069
UCI Website Phishing (variant)	Request_URL	0.595745	0.506494	0.0496499
UCI Website Phishing (variant)	URL_Length	0.579787	0.660944	0.0105902
UCI Website Phishing (variant)	web_traffic	0.569149	0.646288	0.0622996
UCI Website Phishing (variant)	age_of_domain	0.569149	0.630137	0.0527354
UCI Website Phishing (variant)	having_IP_Address	0.56383	0.721088	0

Table T4. Progressive removal on PhiUSIIL: held-out performance after greedily removing the highest-attributed feature at each iteration.

```
[record 1 of 15]
  iteration : 0
  n_features_remaining : 50
  removed_feature : URLSimilarityIndex
  accuracy : 1
  f1 : 1
  mcc : 1
  top_feature_after : LineOfCode
  removed_so_far : 0
removed_features_so_far :
top_feature_before : URLSimilarityIndex
top_shap_before : 0.247807
second_feature : LineOfCode
second_shap : 0.102313
fit_seconds : 21.82
shap_seconds : 0.01
hyperparameter_regime : retuned_per_feature_set
regime_note : hyperparameters tuned once on the full 50-feature set and held fixed; rows after iteration 0
               therefore understate their feature set

[record 2 of 15]
  iteration : 1
  n_features_remaining : 49
  removed_feature : LineOfCode
  accuracy : 0.999716
  f1 : 0.999667
  mcc : 0.99942
  top_feature_after : IsHTTPS
  removed_so_far : 1
removed_features_so_far : URLSimilarityIndex
top_feature_before : LineOfCode
top_shap_before : 0.115271
second_feature : IsHTTPS
second_shap : 0.109668
fit_seconds : 30.77
shap_seconds : 0.01
hyperparameter_regime : tuned_on_full_feature_set
regime_note : hyperparameters tuned once on the full 50-feature set and held fixed; rows after iteration 0
               therefore understate their feature set

[record 3 of 15]
  iteration : 2
  n_features_remaining : 48
  removed_feature : IsHTTPS
  accuracy : 0.999546
  f1 : 0.999467
  mcc : 0.999072
  top_feature_after : NoOfSelfRef
  removed_so_far : 2
removed_features_so_far : URLSimilarityIndex; LineOfCode
top_feature_before : IsHTTPS
top_shap_before : 0.126554
second_feature : NoOfSelfRef
second_shap : 0.0625032
fit_seconds : 28.21
shap_seconds : 0.01
hyperparameter_regime : tuned_on_full_feature_set
regime_note : hyperparameters tuned once on the full 50-feature set and held fixed; rows after iteration 0
               therefore understate their feature set

[record 4 of 15]
  iteration : 3
```

Table T4. Progressive removal on PhiUSIIL: held-out performance after greedily removing the highest-attributed feature at each iteration. (continued)

```

n_features_remaining : 47
  removed_feature : NoOfSelfRef
    accuracy : 0.998156
    f1 : 0.997834
    mcc : 0.99623
  top_feature_after : NoOfExternalRef
  removed_so_far : 3
removed_features_so_far : URLSimilarityIndex; LineOfCode; IsHTTPS
  top_feature_before : NoOfSelfRef
  top_shap_before : 0.106359
  second_feature : NoOfImage
  second_shap : 0.0736897
  fit_seconds : 29.99
  shap_seconds : 0.01
hyperparameter_regime : tuned_on_full_feature_set
  regime_note : hyperparameters tuned once on the full 50-feature set and held fixed; rows after iteration 0
                therefore understate their feature set

[record 5 of 15]
  iteration : 4
n_features_remaining : 46
  removed_feature : NoOfExternalRef
    accuracy : 0.997844
    f1 : 0.997466
    mcc : 0.995594
  top_feature_after : NoOfImage
  removed_so_far : 4
removed_features_so_far : URLSimilarityIndex; LineOfCode; IsHTTPS; NoOfSelfRef
  top_feature_before : NoOfExternalRef
  top_shap_before : 0.0922275
  second_feature : NoOfImage
  second_shap : 0.0850858
  fit_seconds : 30.4
  shap_seconds : 0.01
hyperparameter_regime : tuned_on_full_feature_set
  regime_note : hyperparameters tuned once on the full 50-feature set and held fixed; rows after iteration 0
                therefore understate their feature set

[record 6 of 15]
  iteration : 5
n_features_remaining : 45
  removed_feature : NoOfImage
    accuracy : 0.99773
    f1 : 0.997333
    mcc : 0.995361
  top_feature_after : HasSocialNet
  removed_so_far : 5
removed_features_so_far : URLSimilarityIndex; LineOfCode; IsHTTPS; NoOfSelfRef; NoOfExternalRef
  top_feature_before : NoOfImage
  top_shap_before : 0.103019
  second_feature : HasSocialNet
  second_shap : 0.0522758
  fit_seconds : 30.87
  shap_seconds : 0.01
hyperparameter_regime : tuned_on_full_feature_set
  regime_note : hyperparameters tuned once on the full 50-feature set and held fixed; rows after iteration 0
                therefore understate their feature set

[record 7 of 15]
  iteration : 6
n_features_remaining : 44
  removed_feature : HasSocialNet

```

Table T4. Progressive removal on PhiUS11: held-out performance after greedily removing the highest-attributed feature at each iteration. (continued)

```

        accuracy : 0.997362
          f1 : 0.996899
          mcc : 0.994608
    top_feature_after : NoOfJS
      removed_so_far : 6
removed_features_so_far : URLSimilarityIndex; LineOfCode; IsHTTPS; NoOfSelfRef; NoOfExternalRef; NoOfImage
    top_feature_before : HasSocialNet
    top_shap_before : 0.0748647
      second_feature : NoOfJS
        second_shap : 0.0727016
        fit_seconds : 30.32
        shap_seconds : 0.01
hyperparameter_regime : tuned_on_full_feature_set
  regime_note : hyperparameters tuned once on the full 50-feature set and held fixed; rows after iteration 0
                 therefore understate their feature set

[record 8 of 15]
      iteration : 7
    n_features_remaining : 43
      removed_feature : NoOfJS
        accuracy : 0.996993
          f1 : 0.996465
          mcc : 0.993853
    top_feature_after : NoOfCSS
      removed_so_far : 7
removed_features_so_far : URLSimilarityIndex; LineOfCode; IsHTTPS; NoOfSelfRef; NoOfExternalRef; NoOfImage;
                        HasSocialNet
    top_feature_before : NoOfJS
    top_shap_before : 0.0854281
      second_feature : NoOfOtherSpecialCharsInURL
        second_shap : 0.0511721
        fit_seconds : 32.01
        shap_seconds : 0.01
hyperparameter_regime : tuned_on_full_feature_set
  regime_note : hyperparameters tuned once on the full 50-feature set and held fixed; rows after iteration 0
                 therefore understate their feature set

[record 9 of 15]
      iteration : 8
    n_features_remaining : 42
      removed_feature : NoOfCSS
        accuracy : 0.996567
          f1 : 0.995962
          mcc : 0.992986
    top_feature_after : LargestLineLength
      removed_so_far : 8
removed_features_so_far : URLSimilarityIndex; LineOfCode; IsHTTPS; NoOfSelfRef; NoOfExternalRef; NoOfImage;
                        HasSocialNet; NoOfJS
    top_feature_before : NoOfCSS
    top_shap_before : 0.0897294
      second_feature : NoOfOtherSpecialCharsInURL
        second_shap : 0.0529016
        fit_seconds : 27.76
        shap_seconds : 0.01
hyperparameter_regime : tuned_on_full_feature_set
  regime_note : hyperparameters tuned once on the full 50-feature set and held fixed; rows after iteration 0
                 therefore understate their feature set

[record 10 of 15]
      iteration : 9
    n_features_remaining : 41
      removed_feature : LargestLineLength

```

Table T4. Progressive removal on PhiUSIIL: held-out performance after greedily removing the highest-attributed feature at each iteration. (continued)

```

        accuracy : 0.995376
          f1 : 0.994561
          mcc : 0.990548
    top_feature_after : LetterRatioInURL
      removed_so_far : 9
removed_features_so_far : URLSimilarityIndex; LineOfCode; IsHTTPS; NoOfSelfRef; NoOfExternalRef; NoOfImage;
                        HasSocialNet; NoOfJS; NoOfCSS
    top_feature_before : LargestLineLength
      top_shap_before : 0.0607344
      second_feature : LetterRatioInURL
        second_shap : 0.0535199
        fit_seconds : 23.92
        shap_seconds : 0.01
hyperparameter_regime : tuned_on_full_feature_set
  regime_note : hyperparameters tuned once on the full 50-feature set and held fixed; rows after iteration 0
                  therefore understate their feature set

[record 11 of 15]
      iteration : 10
    n_features_remaining : 40
      removed_feature : LetterRatioInURL
        accuracy : 0.994581
          f1 : 0.993624
          mcc : 0.988925
    top_feature_after : NoOfOtherSpecialCharsInURL
      removed_so_far : 10
removed_features_so_far : URLSimilarityIndex; LineOfCode; IsHTTPS; NoOfSelfRef; NoOfExternalRef; NoOfImage;
                        HasSocialNet; NoOfJS; NoOfCSS; LargestLineLength
    top_feature_before : LetterRatioInURL
      top_shap_before : 0.0583809
      second_feature : NoOfOtherSpecialCharsInURL
        second_shap : 0.0546026
        fit_seconds : 23.86
        shap_seconds : 0.01
hyperparameter_regime : tuned_on_full_feature_set
  regime_note : hyperparameters tuned once on the full 50-feature set and held fixed; rows after iteration 0
                  therefore understate their feature set

[record 12 of 15]
      iteration : 11
    n_features_remaining : 39
      removed_feature : NoOfOtherSpecialCharsInURL
        accuracy : 0.992766
          f1 : 0.991483
          mcc : 0.985213
    top_feature_after : NoOfLettersInURL
      removed_so_far : 11
removed_features_so_far : URLSimilarityIndex; LineOfCode; IsHTTPS; NoOfSelfRef; NoOfExternalRef; NoOfImage;
                        HasSocialNet; NoOfJS; NoOfCSS; LargestLineLength; LetterRatioInURL
    top_feature_before : NoOfOtherSpecialCharsInURL
      top_shap_before : 0.0654809
      second_feature : HasCopyrightInfo
        second_shap : 0.0521448
        fit_seconds : 24.34
        shap_seconds : 1718.39
hyperparameter_regime : tuned_on_full_feature_set
  regime_note : hyperparameters tuned once on the full 50-feature set and held fixed; rows after iteration 0
                  therefore understate their feature set

[record 13 of 15]
      iteration : 12
    n_features_remaining : 38

```

Table T4. Progressive removal on PhiUSIIL: held-out performance after greedily removing the highest-attributed feature at each iteration. (continued)

```

    removed_feature : NoOfLettersInURL
    accuracy : 0.992567
    f1 : 0.991251
    mcc : 0.984805
    top_feature_after : HasCopyrightInfo
    removed_so_far : 12
removed_features_so_far : URLSimilarityIndex; LineOfCode; IsHTTPS; NoOfSelfRef; NoOfExternalRef; NoOfImage;
    HasSocialNet; NoOfJS; NoOfCSS; LargestLineLength; LetterRatioInURL;
    NoOfOtherSpecialCharsInURL
    top_feature_before : NoOfLettersInURL
    top_shap_before : 0.0557819
    second_feature : HasCopyrightInfo
    second_shap : 0.0509389
    fit_seconds : 31.67
    shap_seconds : 1597.42
hyperparameter_regime : tuned_on_full_feature_set
    regime_note : hyperparameters tuned once on the full 50-feature set and held fixed; rows after iteration 0
                    therefore understate their feature set

[record 14 of 15]
    iteration : 13
    n_features_remaining : 37
    removed_feature : HasCopyrightInfo
    accuracy : 0.991801
    f1 : 0.990351
    mcc : 0.983237
    top_feature_after : HasDescription
    removed_so_far : 13
removed_features_so_far : URLSimilarityIndex; LineOfCode; IsHTTPS; NoOfSelfRef; NoOfExternalRef; NoOfImage;
    HasSocialNet; NoOfJS; NoOfCSS; LargestLineLength; LetterRatioInURL;
    NoOfOtherSpecialCharsInURL; NoOfLettersInURL
    top_feature_before : HasCopyrightInfo
    top_shap_before : 0.0573962
    second_feature : HasDescription
    second_shap : 0.0535062
    fit_seconds : 24.72
    shap_seconds : 1571.49
hyperparameter_regime : tuned_on_full_feature_set
    regime_note : hyperparameters tuned once on the full 50-feature set and held fixed; rows after iteration 0
                    therefore understate their feature set

[record 15 of 15]
    iteration : 14
    n_features_remaining : 36
    removed_feature :
    accuracy : 0.989191
    f1 : 0.987306
    mcc : 0.977896
    top_feature_after :
    removed_so_far : 14
removed_features_so_far : URLSimilarityIndex; LineOfCode; IsHTTPS; NoOfSelfRef; NoOfExternalRef; NoOfImage;
    HasSocialNet; NoOfJS; NoOfCSS; LargestLineLength; LetterRatioInURL;
    NoOfOtherSpecialCharsInURL; NoOfLettersInURL; HasCopyrightInfo
    top_feature_before : HasDescription
    top_shap_before : 0.0594322
    second_feature : DomainTitleMatchScore
    second_shap : 0.0528102
    fit_seconds : 33
    shap_seconds : 2022.07
hyperparameter_regime : tuned_on_full_feature_set
    regime_note : hyperparameters tuned once on the full 50-feature set and held fixed; rows after iteration 0
                    therefore understate their feature set

```

Table T4. Progressive removal on PhiUSIL: held-out performance after greedily removing the highest-attributed feature at each iteration. (continued)

Table T6. Zero-shot transfer between benchmarks on the shared nine-concept schema, five seeds per direction.

```
[record 1 of 4]
    direction : PhiUSIIL -> PhiUSIIL
      train : PhiUSIIL
      test : PhiUSIIL
    in_distribution : True
      n_seeds : 5
      n_test : 35249
majority_class_accuracy : 0.57386
  accuracy_mean : 0.901926
  accuracy_std : 0.00065986
accuracy_ci_lower : 0.901393
accuracy_ci_upper : 0.902437
  precision_mean : 0.905644
  precision_std : 0.00163843
precision_ci_lower : 0.904358
precision_ci_upper : 0.906835
  recall_mean : 0.859397
  recall_std : 0.00178201
recall_ci_lower : 0.858025
recall_ci_upper : 0.86077
  f1_mean : 0.881913
  f1_std : 0.000811863
f1_ci_lower : 0.881308
f1_ci_upper : 0.882589
  mcc_mean : 0.798955
  mcc_std : 0.00135952
mcc_ci_lower : 0.797848
mcc_ci_upper : 0.800012
  roc_auc_mean : 0.966993
  roc_auc_std : 0.000351251
roc_auc_ci_lower : 0.966729
roc_auc_ci_upper : 0.967257
accuracy_above_majority : 0.328066

[record 2 of 4]
    direction : PhiUSIIL -> UCI Phishing Websites
      train : PhiUSIIL
      test : UCI Phishing Websites
    in_distribution : False
      n_seeds : 5
      n_test : 1659
majority_class_accuracy : 0.556962
  accuracy_mean : 0.526582
  accuracy_std : 0.00517824
accuracy_ci_lower : 0.522604
accuracy_ci_upper : 0.531046
  precision_mean : 0.483122
  precision_std : 0.00276208
precision_ci_lower : 0.481002
precision_ci_upper : 0.485518
  recall_mean : 0.980408
  recall_std : 0.00354787
recall_ci_lower : 0.977952
recall_ci_upper : 0.983401
  f1_mean : 0.647272
  f1_std : 0.00222726
f1_ci_lower : 0.645507
f1_ci_upper : 0.649001
  mcc_mean : 0.240783
  mcc_std : 0.00866349
mcc_ci_lower : 0.233739
mcc_ci_upper : 0.247386
```

Table T6. Zero-shot transfer between benchmarks on the shared nine-concept schema, five seeds per direction. (continued)

```

roc_auc_mean : 0.840196
roc_auc_std : 0.0256362
roc_auc_ci_lower : 0.818209
roc_auc_ci_upper : 0.855561
accuracy_above_majority : -0.0303797

```

[record 3 of 4]

```

direction : UCI Phishing Websites -> UCI Phishing Websites
train : UCI Phishing Websites
test : UCI Phishing Websites
in_distribution : True
n_seeds : 5
n_test : 1659
majority_class_accuracy : 0.556962
accuracy_mean : 0.920555
accuracy_std : 0.00551463
accuracy_ci_lower : 0.916576
accuracy_ci_upper : 0.925015
precision_mean : 0.928436
precision_std : 0.00632311
precision_ci_lower : 0.923213
precision_ci_upper : 0.933204
recall_mean : 0.889252
recall_std : 0.010565
recall_ci_lower : 0.880272
recall_ci_upper : 0.896599
f1_mean : 0.908391
f1_std : 0.006651
f1_ci_lower : 0.903405
f1_ci_upper : 0.913759
mcc_mean : 0.838964
mcc_std : 0.0111232
mcc_ci_lower : 0.830945
mcc_ci_upper : 0.847961
roc_auc_mean : 0.974095
roc_auc_std : 0.00227702
roc_auc_ci_lower : 0.97243
roc_auc_ci_upper : 0.975889
accuracy_above_majority : 0.363593

```

[record 4 of 4]

```

direction : UCI Phishing Websites -> PhiUSIIL
train : UCI Phishing Websites
test : PhiUSIIL
in_distribution : False
n_seeds : 5
n_test : 35249
majority_class_accuracy : 0.57386
accuracy_mean : 0.640205
accuracy_std : 0.0113818
accuracy_ci_lower : 0.630043
accuracy_ci_upper : 0.646816
precision_mean : 0.859369
precision_std : 0.0165619
precision_ci_lower : 0.846818
precision_ci_upper : 0.872978
recall_mean : 0.185913
recall_std : 0.0298116
recall_ci_lower : 0.159883
recall_ci_upper : 0.203701
f1_mean : 0.30483
f1_std : 0.0420245

```

Table T6. Zero-shot transfer between benchmarks on the shared nine-concept schema, five seeds per direction. (continued)

```
f1_ci_lower : 0.267862
f1_ci_upper : 0.329377
  mcc_mean : 0.278765
  mcc_std : 0.0297142
mcc_ci_lower : 0.252124
mcc_ci_upper : 0.296169
  roc_auc_mean : 0.714418
  roc_auc_std : 0.0161339
roc_auc_ci_lower : 0.704083
roc_auc_ci_upper : 0.728376
accuracy_above_majority : 0.0663451
```

Table T7. Within-dataset ceiling: Kendall's tau between attribution rankings from independent models on disjoint halves of the same data.

[record 1 of 4]

```
config : PhiUSIIL
dataset : PhiUSIIL
n_seeds : 5
n_features : 50
top1_agreement_rate : 1
n_features_sign_top10_mean : 9.2
kendall_tau_mean : 0.91902
kendall_tau_std : 0.0175311
kendall_tau_ci_lower : 0.905306
kendall_tau_ci_upper : 0.932735
jaccard_5_mean : 0.933333
jaccard_5_std : 0.149071
jaccard_5_ci_lower : 0.8
jaccard_5_ci_upper : 1
jaccard_10_mean : 0.860606
jaccard_10_std : 0.141486
jaccard_10_ci_lower : 0.757576
jaccard_10_ci_upper : 0.963636
jaccard_15_mean : 0.852941
jaccard_15_std : 0.049325
jaccard_15_ci_lower : 0.808824
jaccard_15_ci_upper : 0.875
sign_agreement_top5_mean : 34
sign_agreement_top5_std : 24.0832
sign_agreement_top5_ci_lower : 12
sign_agreement_top5_ci_upper : 52
sign_agreement_top10_mean : 41.2222
sign_agreement_top10_std : 19.0289
sign_agreement_top10_ci_lower : 26.8667
sign_agreement_top10_ci_upper : 55.5833
sign_agreement_top15_mean : 41.8681
sign_agreement_top15_std : 13.2519
sign_agreement_top15_ci_lower : 33.7363
sign_agreement_top15_ci_upper : 54.2857
sign_agreement_weighted_mean : 26.2499
sign_agreement_weighted_std : 30.1886
sign_agreement_weighted_ci_lower : 5.24999
sign_agreement_weighted_ci_upper : 52.8187
sign_agreement_unfiltered_mean : 43.6
sign_agreement_unfiltered_std : 11.2606
sign_agreement_unfiltered_ci_lower : 36
sign_agreement_unfiltered_ci_upper : 53.2
```

[record 2 of 4]

```
config : PhiUSIIL (leak-controlled)
dataset : PhiUSIIL
n_seeds : 5
n_features : 36
top1_agreement_rate : 0.8
n_features_sign_top10_mean : 9.6
kendall_tau_mean : 0.930794
kendall_tau_std : 0.00988725
kendall_tau_ci_lower : 0.92254
kendall_tau_ci_upper : 0.937143
jaccard_5_mean : 0.866667
jaccard_5_std : 0.182574
jaccard_5_ci_lower : 0.733333
jaccard_5_ci_upper : 1
jaccard_10_mean : 0.927273
jaccard_10_std : 0.0995859
jaccard_10_ci_lower : 0.854545
```

Table T7. Within-dataset ceiling: Kendall's tau between attribution rankings from independent models on disjoint halves of the same data. (continued)

```

jaccard_10_ci_upper : 1
jaccard_15_mean : 0.9
jaccard_15_std : 0.0559017
jaccard_15_ci_lower : 0.875
jaccard_15_ci_upper : 0.95
sign_agreement_top5_mean : 62
sign_agreement_top5_std : 30.3315
sign_agreement_top5_ci_lower : 38
sign_agreement_top5_ci_upper : 84.05
sign_agreement_top10_mean : 51.3333
sign_agreement_top10_std : 21.1753
sign_agreement_top10_ci_lower : 35.3333
sign_agreement_top10_ci_upper : 68
sign_agreement_top15_mean : 44
sign_agreement_top15_std : 16.7941
sign_agreement_top15_ci_lower : 32
sign_agreement_top15_ci_upper : 55.7143
sign_agreement_weighted_mean : 47.1327
sign_agreement_weighted_std : 19.1492
sign_agreement_weighted_ci_lower : 33.0854
sign_agreement_weighted_ci_upper : 61.7377
sign_agreement_unfiltered_mean : 48.8889
sign_agreement_unfiltered_std : 11.7195
sign_agreement_unfiltered_ci_lower : 40.5556
sign_agreement_unfiltered_ci_upper : 58.8889

```

[record 3 of 4]

```

config : UCI Phishing Websites (de-duplicated)
dataset : UCI Phishing Websites
n_seeds : 5
n_features : 30
top1_agreement_rate : 0.4
n_features_sign_top10_mean : 9
kendall_tau_mean : 0.768276
kendall_tau_std : 0.0115401
kendall_tau_ci_lower : 0.75908
kendall_tau_ci_upper : 0.776575
jaccard_5_mean : 0.666667
jaccard_5_std : 0
jaccard_5_ci_lower : 0.666667
jaccard_5_ci_upper : 0.666667
jaccard_10_mean : 0.818182
jaccard_10_std : 1.24127e-16
jaccard_10_ci_lower : 0.818182
jaccard_10_ci_upper : 0.818182
jaccard_15_mean : 0.789216
jaccard_15_std : 0.0879455
jaccard_15_ci_lower : 0.72549
jaccard_15_ci_upper : 0.852941
sign_agreement_top5_mean : 50
sign_agreement_top5_std : 17.6777
sign_agreement_top5_ci_lower : 35
sign_agreement_top5_ci_upper : 65
sign_agreement_top10_mean : 44.4444
sign_agreement_top10_std : 17.5682
sign_agreement_top10_ci_lower : 31.1111
sign_agreement_top10_ci_upper : 57.7778
sign_agreement_top15_mean : 51.685
sign_agreement_top15_std : 12.9094
sign_agreement_top15_ci_lower : 41.7582
sign_agreement_top15_ci_upper : 61.6117
sign_agreement_weighted_mean : 55.6957

```

Table T7. Within-dataset ceiling: Kendall's tau between attribution rankings from independent models on disjoint halves of the same data. (continued)

```
sign_agreement_weighted_std : 13.8679
sign_agreement_weighted_ci_lower : 45.5465
sign_agreement_weighted_ci_upper : 67.4257
sign_agreement_unfiltered_mean : 50.6667
sign_agreement_unfiltered_std : 4.94413
sign_agreement_unfiltered_ci_lower : 46.6667
sign_agreement_unfiltered_ci_upper : 54
```

[record 4 of 4]

```
config : UCI Phishing Websites
dataset : UCI Phishing Websites
n_seeds : 5
n_features : 30
top1_agreement_rate : 0.6
n_features_sign_top10_mean : 9.6
kendall_tau_mean : 0.824368
kendall_tau_std : 0.0185054
kendall_tau_ci_lower : 0.809632
kendall_tau_ci_upper : 0.838161
jaccard_5_mean : 0.8
jaccard_5_std : 0.182574
jaccard_5_ci_lower : 0.666667
jaccard_5_ci_upper : 0.933333
jaccard_10_mean : 0.927273
jaccard_10_std : 0.0995859
jaccard_10_ci_lower : 0.854545
jaccard_10_ci_upper : 1
jaccard_15_mean : 0.9
jaccard_15_std : 0.0559017
jaccard_15_ci_lower : 0.875
jaccard_15_ci_upper : 0.95
sign_agreement_top5_mean : 45
sign_agreement_top5_std : 13.2288
sign_agreement_top5_ci_lower : 33
sign_agreement_top5_ci_upper : 54
sign_agreement_top10_mean : 44
sign_agreement_top10_std : 15.6663
sign_agreement_top10_ci_lower : 32
sign_agreement_top10_ci_upper : 56
sign_agreement_top15_mean : 46.381
sign_agreement_top15_std : 11.2566
sign_agreement_top15_ci_lower : 37.1429
sign_agreement_top15_ci_upper : 54.0952
sign_agreement_weighted_mean : 47.1409
sign_agreement_weighted_std : 24.1626
sign_agreement_weighted_ci_lower : 26.4575
sign_agreement_weighted_ci_upper : 64.9234
sign_agreement_unfiltered_mean : 46
sign_agreement_unfiltered_std : 4.34613
sign_agreement_unfiltered_ci_lower : 42.6667
sign_agreement_unfiltered_ci_upper : 49.3333
```

Table T11. Frozen feature concept mapping between UCI Phishing Websites and PhiUSIIL, with per-concept directional agreement.

```
[record 1 of 69]
  concept : ip_in_domain
  status : mapped
  uci_feature : having_ip_address
  phiusiil_source : IsDomainIP
  rationale : Both encode 'the host is a bare IP address rather than a domain name'. Direct binary correspondence.
  caveat :
  rho_phiusiil : -0.060549
  rho_uci : -0.0941601
direction_agrees : True
polarity_status : agrees

[record 2 of 69]
  concept : url_length
  status : mapped
  uci_feature : url_length
  phiusiil_source : URLLength
  rationale : Same quantity (characters in the URL); PhiUSIIL stores it raw, UCI stores it binned. Discretised with UCI's published 54/75 thresholds.
  caveat :
  rho_phiusiil : -0.353994
  rho_uci : -0.0533316
direction_agrees : True
polarity_status : agrees

[record 3 of 69]
  concept : subdomain_depth
  status : mapped
  uci_feature : having_sub_domain
  phiusiil_source : NoOfSubDomain
  rationale : Both encode subdomain nesting depth. UCI counts dots after stripping www and ccTLD; PhiUSIIL counts subdomains directly, so the counts align up to the stripping convention.
  caveat : UCI's dot-count and PhiUSIIL's subdomain count can differ by one for hosts with a ccTLD (e.g. .co.uk).
  rho_phiusiil : -0.114149
  rho_uci : -0.304692
direction_agrees : True
polarity_status : agrees

[record 4 of 69]
  concept : ssl_state
  status : mapped
  uci_feature : sslfinal_state
  phiusiil_source : IsHTTPS
  rationale : Closest available correspondence to UCI's SSL assessment: whether the page is served over HTTPS at all, which is the dominant component of sslfinal_state.
  caveat : IMPERFECT. UCI's sslfinal_state additionally requires a trusted issuer and a certificate at least a year old, and has a 'suspicious' state for https-with-untrusted-issuer. PhiUSIIL records only the scheme, so the suspicious state is unreachable from the PhiUSIIL side.
  rho_phiusiil : -0.612415
  rho_uci : -0.735814
direction_agrees : True
polarity_status : agrees

[record 5 of 69]
  concept : iframe
  status : mapped
  uci_feature : iframe
  phiusiil_source : NoOfiFrame
  rationale : Both encode iframe usage on the page.
  caveat : UCI targets specifically INVISIBLE iframes; PhiUSIIL counts all iframes. Note also that UCI's
```

Table T11. Frozen feature concept mapping between UCI Phishing Websites and PhiUSIIL, with per-concept directional agreement. (continued)

	iframe feature carries no marginal signal at all (phishing rate 0.44 at both of its values), so this concept contributes nothing on the UCI side regardless of the mapping quality.
rho_phiusiil :	0.521332
rho_uci :	0.00339352
direction_agrees :	False
polarity_status :	NO_SIGNAL_UCI
[record 6 of 69]	
concept :	popup
status :	mapped
uci_feature :	popupwindow
phiusiil_source :	NoOfPopup
rationale :	Both encode popup-window usage.
caveat :	UCI targets popups containing a text field; PhiUSIIL counts all popups. As with iframe, UCI's popupwindow has phishing rate 0.44 at both values - zero marginal signal on the UCI side.
rho_phiusiil :	0.192546
rho_uci :	-8.58868e-05
direction_agrees :	False
polarity_status :	NO_SIGNAL_UCI
[record 7 of 69]	
concept :	url_redirect
status :	mapped
uci_feature :	redirect
phiusiil_source :	NoOfURLRedirect
rationale :	Both encode redirection on the way to the page. UCI's `redirect` takes only {0, 1} in the delivered data, so the PhiUSIIL side is mapped onto that same {0, 1} domain rather than the usual ternary one.
caveat :	UCI's published rule describes a 3-way split on redirect count, but the delivered column is binary; the mapping follows the DATA, not the paper.
rho_phiusiil :	0.0480144
rho_uci :	0.0201135
direction_agrees :	True
polarity_status :	agrees
[record 8 of 69]	
concept :	form_handler
status :	mapped
uci_feature :	sfh
phiusiil_source :	HasExternalFormSubmit
rationale :	Both encode whether a form posts to a different domain than the page - UCI's Server Form Handler check and PhiUSIIL's external-form-submit flag.
caveat :	UCI's 'suspicious' state (blank/about:blank handler) has no PhiUSIIL counterpart and is unreachable from the PhiUSIIL side.
rho_phiusiil :	0.167309
rho_uci :	-0.21919
direction_agrees :	False
polarity_status :	INVERTED
[record 9 of 69]	
concept :	anchor_external_ratio
status :	mapped
uci_feature :	url_of_anchor
phiusiil_source :	NoOfExternalRef + NoOfSelfRef + NoOfEmptyRef
rationale :	UCI's url_of_anchor is the share of <a> hrefs pointing off-domain. PhiUSIIL provides the underlying counts (external / self / empty references), so the ratio is reconstructed directly and binned with UCI's 31%/67% thresholds.
caveat :	
rho_phiusiil :	0.0872799
rho_uci :	-0.701207
direction_agrees :	False
polarity_status :	INVERTED

Table T11. Frozen feature concept mapping between UCI Phishing Websites and PhiUSIIL, with per-concept directional agreement. (continued)

```
[record 10 of 69]
  concept :
    status : unmapped_uci
    uci_feature : shortining_service
  phiusiil_source :
    rationale : URL-shortener detection; PhiUSIIL has no shortener feature.
    caveat :
  rho_phiusiil :
  rho_uci :
direction_agrees :
polarity_status :

[record 11 of 69]
  concept :
    status : unmapped_uci
    uci_feature : having_at_symbol
  phiusiil_source :
    rationale : presence of '@' in the URL; PhiUSIIL counts only aggregate special characters, with no per-
    character breakdown.
    caveat :
  rho_phiusiil :
  rho_uci :
direction_agrees :
polarity_status :

[record 12 of 69]
  concept :
    status : unmapped_uci
    uci_feature : double_slash_redirecting
  phiusiil_source :
    rationale : '//' position in the path; no PhiUSIIL equivalent.
    caveat :
  rho_phiusiil :
  rho_uci :
direction_agrees :
polarity_status :

[record 13 of 69]
  concept :
    status : unmapped_uci
    uci_feature : prefix_suffix
  phiusiil_source :
    rationale : hyphen in the domain; PhiUSIIL has no hyphen feature.
    caveat :
  rho_phiusiil :
  rho_uci :
direction_agrees :
polarity_status :

[record 14 of 69]
  concept :
    status : unmapped_uci
    uci_feature : domain_registration_length
  phiusiil_source :
    rationale : WHOIS registration length; PhiUSIIL has no WHOIS data.
    caveat :
  rho_phiusiil :
  rho_uci :
direction_agrees :
polarity_status :
```

Table T11. Frozen feature concept mapping between UCI Phishing Websites and PhiUSIIL, with per-concept directional agreement. (continued)

```
[record 15 of 69]
  concept :
    status : unmapped_uci
  uci_feature : favicon
  phiusiil_source :
    rationale : favicon loaded from an EXTERNAL domain. PhiUSIIL's HasFavicon records presence only, which is a
                different concept - mapping them would force a match between 'has a favicon' and 'favicon is
                foreign'.
    caveat :
  rho_phiusiil :
  rho_uci :
direction_agrees :
polarity_status :

[record 16 of 69]
  concept :
    status : unmapped_uci
  uci_feature : port
  phiusiil_source :
    rationale : non-standard port; no PhiUSIIL equivalent.
    caveat :
  rho_phiusiil :
  rho_uci :
direction_agrees :
polarity_status :

[record 17 of 69]
  concept :
    status : unmapped_uci
  uci_feature : https_token
  phiusiil_source :
    rationale : the literal token 'https' inside the domain name (a deception pattern), NOT whether HTTPS is used.
                PhiUSIIL's IsHTTPS is the latter and is already mapped to sslfinal_state.
    caveat :
  rho_phiusiil :
  rho_uci :
direction_agrees :
polarity_status :

[record 18 of 69]
  concept :
    status : unmapped_uci
  uci_feature : request_url
  phiusiil_source :
    rationale : share of loaded OBJECTS (images/scripts) from an external domain. PhiUSIIL has
                NoOfImage/NoOfCSS/NoOfJS but no internal/external split for them, so the ratio cannot be
                reconstructed.
    caveat :
  rho_phiusiil :
  rho_uci :
direction_agrees :
polarity_status :

[record 19 of 69]
  concept :
    status : unmapped_uci
  uci_feature : links_in_tags
  phiusiil_source :
    rationale : share of links inside meta/script/link tags; PhiUSIIL has no tag-scoped link counts.
    caveat :
  rho_phiusiil :
  rho_uci :
```

Table T11. Frozen feature concept mapping between UCI Phishing Websites and PhiUSIIL, with per-concept directional agreement. (continued)

```

direction_agrees :
polarity_status :

[record 20 of 69]
  concept :
    status : unmapped_uci
  uci_feature : submitting_to_email
  phiusiil_source :
    rationale : mailto: form action; no PhiUSIIL equivalent.
    caveat :
  rho_phiusiil :
  rho_uci :
direction_agrees :
polarity_status :

[record 21 of 69]
  concept :
    status : unmapped_uci
  uci_feature : abnormal_url
  phiusiil_source :
    rationale : WHOIS hostname consistency; no PhiUSIIL WHOIS data.
    caveat :
  rho_phiusiil :
  rho_uci :
direction_agrees :
polarity_status :

[record 22 of 69]
  concept :
    status : unmapped_uci
  uci_feature : on_mouseover
  phiusiil_source :
    rationale : JavaScript status-bar manipulation; PhiUSIIL records no JS behaviour.
    caveat :
  rho_phiusiil :
  rho_uci :
direction_agrees :
polarity_status :

[record 23 of 69]
  concept :
    status : unmapped_uci
  uci_feature : rightclick
  phiusiil_source :
    rationale : right-click disabling; PhiUSIIL records no JS behaviour.
    caveat :
  rho_phiusiil :
  rho_uci :
direction_agrees :
polarity_status :

[record 24 of 69]
  concept :
    status : unmapped_uci
  uci_feature : age_of_domain
  phiusiil_source :
    rationale : WHOIS domain age; no PhiUSIIL WHOIS data.
    caveat :
  rho_phiusiil :
  rho_uci :
direction_agrees :
polarity_status :

```

Table T11. Frozen feature concept mapping between UCI Phishing Websites and PhiUSIIL, with per-concept directional agreement. (continued)

```
[record 25 of 69]
  concept :
    status : unmapped_uci
    uci_feature : dnsrecord
  phiusiil_source :
    rationale : DNS record existence; no PhiUSIIL DNS data.
    caveat :
    rho_phiusiil :
    rho_uci :
direction_agrees :
polarity_status :

[record 26 of 69]
  concept :
    status : unmapped_uci
    uci_feature : web_traffic
  phiusiil_source :
    rationale : Alexa traffic rank; external reputation, absent from PhiUSIIL.
    caveat :
    rho_phiusiil :
    rho_uci :
direction_agrees :
polarity_status :

[record 27 of 69]
  concept :
    status : unmapped_uci
    uci_feature : page_rank
  phiusiil_source :
    rationale : PageRank; external reputation, absent from PhiUSIIL.
    caveat :
    rho_phiusiil :
    rho_uci :
direction_agrees :
polarity_status :

[record 28 of 69]
  concept :
    status : unmapped_uci
    uci_feature : google_index
  phiusiil_source :
    rationale : Google index membership; external reputation, absent from PhiUSIIL.
    caveat :
    rho_phiusiil :
    rho_uci :
direction_agrees :
polarity_status :

[record 29 of 69]
  concept :
    status : unmapped_uci
    uci_feature : links_pointing_to_page
  phiusiil_source :
    rationale : INBOUND link count. PhiUSIIL's NoOfExternalRef is OUTBOUND; mapping them would invert the direction
    of the relationship.
    caveat :
    rho_phiusiil :
    rho_uci :
direction_agrees :
polarity_status :
```

Table T11. Frozen feature concept mapping between UCI Phishing Websites and PhiUSIIL, with per-concept directional agreement. (continued)

```
[record 30 of 69]
  concept :
    status : unmapped_uci
  uci_feature : statistical_report
  phiusiil_source :
    rationale : membership of a published blacklist; external reputation.
    caveat :
  rho_phiusiil :
    rho_uci :
direction_agrees :
polarity_status :

[record 31 of 69]
  concept :
    status : unmapped_phiusiil
  uci_feature :
  phiusiil_source : DomainLength
    rationale : no defensible UCI counterpart (UCI has no comparable page-content/URL-composition feature)
    caveat :
  rho_phiusiil :
    rho_uci :
direction_agrees :
polarity_status :

[record 32 of 69]
  concept :
    status : unmapped_phiusiil
  uci_feature :
  phiusiil_source : URLSimilarityIndex
    rationale : no defensible UCI counterpart (UCI has no comparable page-content/URL-composition feature)
    caveat :
  rho_phiusiil :
    rho_uci :
direction_agrees :
polarity_status :

[record 33 of 69]
  concept :
    status : unmapped_phiusiil
  uci_feature :
  phiusiil_source : CharContinuationRate
    rationale : no defensible UCI counterpart (UCI has no comparable page-content/URL-composition feature)
    caveat :
  rho_phiusiil :
    rho_uci :
direction_agrees :
polarity_status :

[record 34 of 69]
  concept :
    status : unmapped_phiusiil
  uci_feature :
  phiusiil_source : TLDLegitimateProb
    rationale : no defensible UCI counterpart (UCI has no comparable page-content/URL-composition feature)
    caveat :
  rho_phiusiil :
    rho_uci :
direction_agrees :
polarity_status :

[record 35 of 69]
  concept :
```

Table T11. Frozen feature concept mapping between UCI Phishing Websites and PhiUSIIL, with per-concept directional agreement. (continued)

```

    status : unmapped_phiusiil
    uci_feature :
    phiusiil_source : URLCharProb
    rationale : no defensible UCI counterpart (UCI has no comparable page-content/URL-composition feature)
    caveat :
    rho_phiusiil :
    rho_uci :
direction_agrees :
polarity_status :

[record 36 of 69]
    concept :
    status : unmapped_phiusiil
    uci_feature :
    phiusiil_source : TLDLength
    rationale : no defensible UCI counterpart (UCI has no comparable page-content/URL-composition feature)
    caveat :
    rho_phiusiil :
    rho_uci :
direction_agrees :
polarity_status :

[record 37 of 69]
    concept :
    status : unmapped_phiusiil
    uci_feature :
    phiusiil_source : HasObfuscation
    rationale : no defensible UCI counterpart (UCI has no comparable page-content/URL-composition feature)
    caveat :
    rho_phiusiil :
    rho_uci :
direction_agrees :
polarity_status :

[record 38 of 69]
    concept :
    status : unmapped_phiusiil
    uci_feature :
    phiusiil_source : NoOf0bfuscatedChar
    rationale : no defensible UCI counterpart (UCI has no comparable page-content/URL-composition feature)
    caveat :
    rho_phiusiil :
    rho_uci :
direction_agrees :
polarity_status :

[record 39 of 69]
    concept :
    status : unmapped_phiusiil
    uci_feature :
    phiusiil_source : ObfuscationRatio
    rationale : no defensible UCI counterpart (UCI has no comparable page-content/URL-composition feature)
    caveat :
    rho_phiusiil :
    rho_uci :
direction_agrees :
polarity_status :

[record 40 of 69]
    concept :
    status : unmapped_phiusiil
    uci_feature :

```

Table T11. Frozen feature concept mapping between UCI Phishing Websites and PhiUSIIL, with per-concept directional agreement. (continued)

```

phiusiil_source : NoOfLettersInURL
  rationale : no defensible UCI counterpart (UCI has no comparable page-content/URL-composition feature)
  caveat :
rho_phiusiil :
rho_uci :
direction_agrees :
polarity_status :

[record 41 of 69]
  concept :
    status : unmapped_phiusiil
  uci_feature :
phiusiil_source : LetterRatioInURL
  rationale : no defensible UCI counterpart (UCI has no comparable page-content/URL-composition feature)
  caveat :
rho_phiusiil :
rho_uci :
direction_agrees :
polarity_status :

[record 42 of 69]
  concept :
    status : unmapped_phiusiil
  uci_feature :
phiusiil_source : NoOfDegitsInURL
  rationale : no defensible UCI counterpart (UCI has no comparable page-content/URL-composition feature)
  caveat :
rho_phiusiil :
rho_uci :
direction_agrees :
polarity_status :

[record 43 of 69]
  concept :
    status : unmapped_phiusiil
  uci_feature :
phiusiil_source : DgitRatioInURL
  rationale : no defensible UCI counterpart (UCI has no comparable page-content/URL-composition feature)
  caveat :
rho_phiusiil :
rho_uci :
direction_agrees :
polarity_status :

[record 44 of 69]
  concept :
    status : unmapped_phiusiil
  uci_feature :
phiusiil_source : NoOfEqualsInURL
  rationale : no defensible UCI counterpart (UCI has no comparable page-content/URL-composition feature)
  caveat :
rho_phiusiil :
rho_uci :
direction_agrees :
polarity_status :

[record 45 of 69]
  concept :
    status : unmapped_phiusiil
  uci_feature :
phiusiil_source : NoOfQMarkInURL
  rationale : no defensible UCI counterpart (UCI has no comparable page-content/URL-composition feature)

```

Table T11. Frozen feature concept mapping between UCI Phishing Websites and PhiUSIIL, with per-concept directional agreement. (continued)

```

    caveat :
    rho_phiusiil :
    rho_uci :
direction_agrees :
polarity_status :

[record 46 of 69]
    concept :
    status : unmapped_phiusiil
    uci_feature :
    phiusiil_source : NoOfAmpersandInURL
    rationale : no defensible UCI counterpart (UCI has no comparable page-content/URL-composition feature)
    caveat :
    rho_phiusiil :
    rho_uci :
direction_agrees :
polarity_status :

[record 47 of 69]
    concept :
    status : unmapped_phiusiil
    uci_feature :
    phiusiil_source : NoOfOtherSpecialCharsInURL
    rationale : no defensible UCI counterpart (UCI has no comparable page-content/URL-composition feature)
    caveat :
    rho_phiusiil :
    rho_uci :
direction_agrees :
polarity_status :

[record 48 of 69]
    concept :
    status : unmapped_phiusiil
    uci_feature :
    phiusiil_source : SpacialCharRatioInURL
    rationale : no defensible UCI counterpart (UCI has no comparable page-content/URL-composition feature)
    caveat :
    rho_phiusiil :
    rho_uci :
direction_agrees :
polarity_status :

[record 49 of 69]
    concept :
    status : unmapped_phiusiil
    uci_feature :
    phiusiil_source : LineOfCode
    rationale : no defensible UCI counterpart (UCI has no comparable page-content/URL-composition feature)
    caveat :
    rho_phiusiil :
    rho_uci :
direction_agrees :
polarity_status :

[record 50 of 69]
    concept :
    status : unmapped_phiusiil
    uci_feature :
    phiusiil_source : LargestLineLength
    rationale : no defensible UCI counterpart (UCI has no comparable page-content/URL-composition feature)
    caveat :
    rho_phiusiil :

```

Table T11. Frozen feature concept mapping between UCI Phishing Websites and PhiUSIIL, with per-concept directional agreement. (continued)

```

    rho_uci :
direction_agrees :
polarity_status :

[record 51 of 69]
    concept :
    status : unmapped_phiusiil
    uci_feature :
    phiusiil_source : HasTitle
    rationale : no defensible UCI counterpart (UCI has no comparable page-content/URL-composition feature)
    caveat :
    rho_phiusiil :
    rho_uci :
direction_agrees :
polarity_status :

[record 52 of 69]
    concept :
    status : unmapped_phiusiil
    uci_feature :
    phiusiil_source : DomainTitleMatchScore
    rationale : no defensible UCI counterpart (UCI has no comparable page-content/URL-composition feature)
    caveat :
    rho_phiusiil :
    rho_uci :
direction_agrees :
polarity_status :

[record 53 of 69]
    concept :
    status : unmapped_phiusiil
    uci_feature :
    phiusiil_source : URLTitleMatchScore
    rationale : no defensible UCI counterpart (UCI has no comparable page-content/URL-composition feature)
    caveat :
    rho_phiusiil :
    rho_uci :
direction_agrees :
polarity_status :

[record 54 of 69]
    concept :
    status : unmapped_phiusiil
    uci_feature :
    phiusiil_source : HasFavicon
    rationale : no defensible UCI counterpart (UCI has no comparable page-content/URL-composition feature)
    caveat :
    rho_phiusiil :
    rho_uci :
direction_agrees :
polarity_status :

[record 55 of 69]
    concept :
    status : unmapped_phiusiil
    uci_feature :
    phiusiil_source : Robots
    rationale : no defensible UCI counterpart (UCI has no comparable page-content/URL-composition feature)
    caveat :
    rho_phiusiil :
    rho_uci :
direction_agrees :

```

Table T11. Frozen feature concept mapping between UCI Phishing Websites and PhiUSIIL, with per-concept directional agreement. (continued)

```

polarity_status :

[record 56 of 69]
  concept :
    status : unmapped_phiusiil
  uci_feature :
  phiusiil_source : IsResponsive
    rationale : no defensible UCI counterpart (UCI has no comparable page-content/URL-composition feature)
    caveat :
  rho_phiusiil :
  rho_uci :
direction_agrees :
polarity_status :

[record 57 of 69]
  concept :
    status : unmapped_phiusiil
  uci_feature :
  phiusiil_source : NoOfSelfRedirect
    rationale : no defensible UCI counterpart (UCI has no comparable page-content/URL-composition feature)
    caveat :
  rho_phiusiil :
  rho_uci :
direction_agrees :
polarity_status :

[record 58 of 69]
  concept :
    status : unmapped_phiusiil
  uci_feature :
  phiusiil_source : HasDescription
    rationale : no defensible UCI counterpart (UCI has no comparable page-content/URL-composition feature)
    caveat :
  rho_phiusiil :
  rho_uci :
direction_agrees :
polarity_status :

[record 59 of 69]
  concept :
    status : unmapped_phiusiil
  uci_feature :
  phiusiil_source : HasSocialNet
    rationale : no defensible UCI counterpart (UCI has no comparable page-content/URL-composition feature)
    caveat :
  rho_phiusiil :
  rho_uci :
direction_agrees :
polarity_status :

[record 60 of 69]
  concept :
    status : unmapped_phiusiil
  uci_feature :
  phiusiil_source : HasSubmitButton
    rationale : no defensible UCI counterpart (UCI has no comparable page-content/URL-composition feature)
    caveat :
  rho_phiusiil :
  rho_uci :
direction_agrees :
polarity_status :

```

Table T11. Frozen feature concept mapping between UCI Phishing Websites and PhiUSIIL, with per-concept directional agreement. (continued)

```
[record 61 of 69]
  concept :
    status : unmapped_phiusiil
  uci_feature :
  phiusiil_source : HasHiddenFields
    rationale : no defensible UCI counterpart (UCI has no comparable page-content/URL-composition feature)
    caveat :
  rho_phiusiil :
    rho_uci :
direction_agrees :
polarity_status :

[record 62 of 69]
  concept :
    status : unmapped_phiusiil
  uci_feature :
  phiusiil_source : HasPasswordField
    rationale : no defensible UCI counterpart (UCI has no comparable page-content/URL-composition feature)
    caveat :
  rho_phiusiil :
    rho_uci :
direction_agrees :
polarity_status :

[record 63 of 69]
  concept :
    status : unmapped_phiusiil
  uci_feature :
  phiusiil_source : Bank
    rationale : no defensible UCI counterpart (UCI has no comparable page-content/URL-composition feature)
    caveat :
  rho_phiusiil :
    rho_uci :
direction_agrees :
polarity_status :

[record 64 of 69]
  concept :
    status : unmapped_phiusiil
  uci_feature :
  phiusiil_source : Pay
    rationale : no defensible UCI counterpart (UCI has no comparable page-content/URL-composition feature)
    caveat :
  rho_phiusiil :
    rho_uci :
direction_agrees :
polarity_status :

[record 65 of 69]
  concept :
    status : unmapped_phiusiil
  uci_feature :
  phiusiil_source : Crypto
    rationale : no defensible UCI counterpart (UCI has no comparable page-content/URL-composition feature)
    caveat :
  rho_phiusiil :
    rho_uci :
direction_agrees :
polarity_status :

[record 66 of 69]
  concept :
```

Table T11. Frozen feature concept mapping between UCI Phishing Websites and PhiUSIIL, with per-concept directional agreement. (continued)

```

    status : unmapped_phiusiil
    uci_feature :
    phiusiil_source : HasCopyrightInfo
    rationale : no defensible UCI counterpart (UCI has no comparable page-content/URL-composition feature)
    caveat :
    rho_phiusiil :
    rho_uci :
direction_agrees :
polarity_status :

[record 67 of 69]
    concept :
    status : unmapped_phiusiil
    uci_feature :
    phiusiil_source : NoOfImage
    rationale : no defensible UCI counterpart (UCI has no comparable page-content/URL-composition feature)
    caveat :
    rho_phiusiil :
    rho_uci :
direction_agrees :
polarity_status :

[record 68 of 69]
    concept :
    status : unmapped_phiusiil
    uci_feature :
    phiusiil_source : NoOfCSS
    rationale : no defensible UCI counterpart (UCI has no comparable page-content/URL-composition feature)
    caveat :
    rho_phiusiil :
    rho_uci :
direction_agrees :
polarity_status :

[record 69 of 69]
    concept :
    status : unmapped_phiusiil
    uci_feature :
    phiusiil_source : NoOfJS
    rationale : no defensible UCI counterpart (UCI has no comparable page-content/URL-composition feature)
    caveat :
    rho_phiusiil :
    rho_uci :
direction_agrees :
polarity_status :

```

Table T12. Explanation cost profile for progressive removal, annotated with whether each timing was taken under competing CPU load.

[record 1 of 24]

```

dataset : PhiUSIIL
iteration : 0
n_features : 50
accuracy : 1
fit_seconds : 20.46
shap_seconds : 284.19
mean_tree_depth : 13.864
total_nodes : 104862
window_start : 10:34:34
window_end : 10:38:27
cache_replay : False
source_log : a13.log
run_date : 2026-09-11
window_reconstructed : True
competing_jobs :
contention_evidence : remeasured_uncontended
sampled_cpu_percent_mean :
sampled_cpu_percent_max :
sampled_n_python_procs_max :
sampled_job_tags :
n_load_samples :
contended : False
cost_analysis_eligible : True
remeasured_clean : True
shap_seconds_original_contended : 179.9
f1 :
mcc :
removed_feature :
top_feature :
hyperparameter_regime :
```

[record 2 of 24]

```

dataset : PhiUSIIL
iteration : 1
n_features : 49
accuracy : 0.999716
fit_seconds : 31.2
shap_seconds : 683.21
mean_tree_depth : 19.158
total_nodes : 281270
window_start : 10:38:27
window_end : 10:46:39
cache_replay : False
source_log : a13.log
run_date : 2026-09-11
window_reconstructed :
competing_jobs :
contention_evidence : remeasured_uncontended
sampled_cpu_percent_mean :
sampled_cpu_percent_max :
sampled_n_python_procs_max :
sampled_job_tags :
n_load_samples :
contended : False
cost_analysis_eligible : True
remeasured_clean : True
shap_seconds_original_contended : 463.5
f1 :
mcc :
removed_feature :
top_feature :
```

Table T12. Explanation cost profile for progressive removal, annotated with whether each timing was taken under competing CPU load. (continued)

```

hyperparameter_regime :
[record 3 of 24]
    dataset : PhiUSIIL
    iteration : 2
    n_features : 48
    accuracy : 0.999546
    fit_seconds : 27.4
    shap_seconds : 788.7
    mean_tree_depth : 20.152
    total_nodes : 367168
    window_start : 10:46:39
    window_end : 10:56:32
    cache_replay : False
    source_log : a13.log
    run_date : 2026-09-11
    window_reconstructed :
    competing_jobs :
    contention_evidence : remeasured_uncontended
    sampled_cpu_percent_mean :
    sampled_cpu_percent_max :
    sampled_n_python_procs_max :
    sampled_job_tags :
    n_load_samples :
    contended : False
    cost_analysis_eligible : True
    remeasured_clean : True
shap_seconds_original_contended : 564.3
    f1 :
    mcc :
    removed_feature :
    top_feature :
hyperparameter_regime :

```

```

[record 4 of 24]
    dataset : PhiUSIIL
    iteration : 3
    n_features : 47
    accuracy : 0.998156
    fit_seconds : 28.77
    shap_seconds : 978.95
    mean_tree_depth : 23.722
    total_nodes : 544400
    window_start : 10:56:32
    window_end : 11:08:45
    cache_replay : False
    source_log : a13.log
    run_date : 2026-09-11
    window_reconstructed :
    competing_jobs :
    contention_evidence : remeasured_uncontended
    sampled_cpu_percent_mean :
    sampled_cpu_percent_max :
    sampled_n_python_procs_max :
    sampled_job_tags :
    n_load_samples :
    contended : False
    cost_analysis_eligible : True
    remeasured_clean : True
shap_seconds_original_contended : 705.8
    f1 :
    mcc :

```

Table T12. Explanation cost profile for progressive removal, annotated with whether each timing was taken under competing CPU load. (continued)

```

        removed_feature :
        top_feature :
hyperparameter_regime :

[record 5 of 24]
        dataset : PhiUSIIL
        iteration : 4
        n_features : 46
        accuracy : 0.997844
        fit_seconds : 27.9
        shap_seconds : 582.7
        mean_tree_depth : 25.126
        total_nodes : 628132
        window_start : 11:08:45
        window_end : 11:18:56
        cache_replay : False
        source_log : a13.log
        run_date : 2026-09-11
        window_reconstructed :
        competing_jobs :
        contention_evidence :
        sampled_cpu_percent_mean :
        sampled_cpu_percent_max :
        sampled_n_python_procs_max :
        sampled_job_tags :
        n_load_samples :
        contended : False
        cost_analysis_eligible : True
        remeasured_clean : False
shap_seconds_original_contended :
        f1 :
        mcc :
        removed_feature :
        top_feature :
hyperparameter_regime :
```

```

[record 6 of 24]
        dataset : PhiUSIIL
        iteration : 5
        n_features : 45
        accuracy : 0.99773
        fit_seconds : 25.2
        shap_seconds : 851.9
        mean_tree_depth : 26.246
        total_nodes : 704950
        window_start : 11:18:56
        window_end : 11:33:34
        cache_replay : False
        source_log : a13.log
        run_date : 2026-09-11
        window_reconstructed :
        competing_jobs :
        contention_evidence :
        sampled_cpu_percent_mean :
        sampled_cpu_percent_max :
        sampled_n_python_procs_max :
        sampled_job_tags :
        n_load_samples :
        contended : False
        cost_analysis_eligible : True
        remeasured_clean : False
shap_seconds_original_contended :
```

Table T12. Explanation cost profile for progressive removal, annotated with whether each timing was taken under competing CPU load. (continued)

```

        f1 :
        mcc :
    removed_feature :
        top_feature :
    hyperparameter_regime :

[record 7 of 24]
        dataset : PhiUSIIL
        iteration : 6
        n_features : 44
        accuracy : 0.997362
        fit_seconds : 25.4
        shap_seconds : 752.1
    mean_tree_depth : 28.38
        total_nodes : 838026
    window_start : 11:33:34
    window_end : 11:46:32
    cache_replay : False
    source_log : a13.log
    run_date : 2026-09-11
    window_reconstructed :
    competing_jobs :
    contention_evidence :
    sampled_cpu_percent_mean :
    sampled_cpu_percent_max :
    sampled_n_python_procs_max :
    sampled_job_tags :
    n_load_samples :
        contended : False
    cost_analysis_eligible : True
    remeasured_clean : False
shap_seconds_original_contended :
    f1 :
    mcc :
    removed_feature :
    top_feature :
    hyperparameter_regime :

```

```

[record 8 of 24]
        dataset : PhiUSIIL
        iteration : 7
        n_features : 43
        accuracy : 0.996993
        fit_seconds : 32.3
        shap_seconds : 700.5
    mean_tree_depth : 28.88
        total_nodes : 952992
    window_start : 11:46:32
    window_end : 11:58:46
    cache_replay : False
    source_log : a13.log
    run_date : 2026-09-11
    window_reconstructed :
    competing_jobs :
    contention_evidence :
    sampled_cpu_percent_mean :
    sampled_cpu_percent_max :
    sampled_n_python_procs_max :
    sampled_job_tags :
    n_load_samples :
        contended : False
    cost_analysis_eligible : True

```

Table T12. Explanation cost profile for progressive removal, annotated with whether each timing was taken under competing CPU load. (continued)

```

remeasured_clean : False
shap_seconds_original_contended :
    f1 :
    mcc :
    removed_feature :
    top_feature :
hyperparameter_regime :

[record 9 of 24]
    dataset : PhiUSIIL
    iteration : 8
    n_features : 42
    accuracy : 0.996567
    fit_seconds : 24.1
    shap_seconds : 717.8
    mean_tree_depth : 31.152
    total_nodes : 1075554
    window_start : 11:58:46
    window_end : 12:11:08
    cache_replay : False
    source_log : a13.log
    run_date : 2026-09-11
    window_reconstructed :
    competing_jobs :
    contention_evidence :
    sampled_cpu_percent_mean :
    sampled_cpu_percent_max :
    sampled_n_python_procs_max :
    sampled_job_tags :
    n_load_samples :
    contended : False
    cost_analysis_eligible : True
    remeasured_clean : False
shap_seconds_original_contended :
    f1 :
    mcc :
    removed_feature :
    top_feature :
hyperparameter_regime :
```

```

[record 10 of 24]
    dataset : PhiUSIIL
    iteration : 9
    n_features : 41
    accuracy : 0.995376
    fit_seconds : 29.1
    shap_seconds : 1220.1
    mean_tree_depth : 33.43
    total_nodes : 1291626
    window_start : 12:11:08
    window_end : 12:31:58
    cache_replay : False
    source_log : a13.log
    run_date : 2026-09-11
    window_reconstructed :
    competing_jobs :
    contention_evidence :
    sampled_cpu_percent_mean :
    sampled_cpu_percent_max :
    sampled_n_python_procs_max :
    sampled_job_tags :
    n_load_samples :
```

Table T12. Explanation cost profile for progressive removal, annotated with whether each timing was taken under competing CPU load. (continued)

```

        contended : False
cost_analysis_eligible : True
    remeasured_clean : False
shap_seconds_original_contended :
    f1 :
    mcc :
    removed_feature :
    top_feature :
hyperparameter_regime :

```

[record 11 of 24]

```

    dataset : PhiUSIIL
    iteration : 10
    n_features : 40
    accuracy : 0.994581
    fit_seconds : 25.4
    shap_seconds : 1264.2
    mean_tree_depth : 36.51
    total_nodes : 1496914
    window_start : 12:31:58
    window_end : 12:53:28
    cache_replay : False
    source_log : a13.log
    run_date : 2026-09-11
    window_reconstructed :
    competing_jobs :
    contention_evidence :
    sampled_cpu_percent_mean :
    sampled_cpu_percent_max :
    sampled_n_python_procs_max :
    sampled_job_tags :
    n_load_samples :
    contended : False
cost_analysis_eligible : True
    remeasured_clean : False
shap_seconds_original_contended :
    f1 :
    mcc :
    removed_feature :
    top_feature :
hyperparameter_regime :

```

[record 12 of 24]

```

    dataset : PhiUSIIL
    iteration : 11
    n_features : 39
    accuracy : 0.992766
    fit_seconds : 28.83
    shap_seconds : 1860.9
    mean_tree_depth : 39.466
    total_nodes : 1718032
    window_start : 14:59:46
    window_end : 15:28:50
    cache_replay : False
    source_log : a13_resume.log
    run_date : 2026-09-11
    window_reconstructed :
    competing_jobs :
    contention_evidence : remeasured_uncontended
    sampled_cpu_percent_mean :
    sampled_cpu_percent_max :
    sampled_n_python_procs_max :

```

Table T12. Explanation cost profile for progressive removal, annotated with whether each timing was taken under competing CPU load. (continued)

```

sampled_job_tags :
  n_load_samples :
    contended : False
cost_analysis_eligible : True
remeasured_clean : True
shap_seconds_original_contended : 1718.4
  f1 :
  mcc :
removed_feature :
  top_feature :
hyperparameter_regime :

[record 13 of 24]
  dataset : PhiUSIIL
  iteration : 12
  n_features : 38
  accuracy : 0.992567
  fit_seconds : 31.7
  shap_seconds : 1597.4
  mean_tree_depth : 39.31
  total_nodes : 1738726
  window_start : 15:28:50
  window_end : 15:56:00
  cache_replay : False
  source_log : a13_resume.log
  run_date : 2026-09-11
window_reconstructed :
  competing_jobs : B1;B2;B2acq
  contention_evidence : phase_a.log+sampler
sampled_cpu_percent_mean : 38.08
sampled_cpu_percent_max : 100
sampled_n_python_procs_max : 53
sampled_job_tags : B2
  n_load_samples : 46
  contended : True
cost_analysis_eligible : False
remeasured_clean : False
shap_seconds_original_contended :
  f1 :
  mcc :
removed_feature :
  top_feature :
hyperparameter_regime :

[record 14 of 24]
  dataset : PhiUSIIL
  iteration : 13
  n_features : 37
  accuracy : 0.991801
  fit_seconds : 24.7
  shap_seconds : 1571.5
  mean_tree_depth : 39.632
  total_nodes : 1866634
  window_start : 15:56:00
  window_end : 16:22:37
  cache_replay : False
  source_log : a13_resume.log
  run_date : 2026-09-11
window_reconstructed :
  competing_jobs : B2;B2acq
  contention_evidence : phase_a.log+sampler
sampled_cpu_percent_mean : 19.24

```

Table T12. Explanation cost profile for progressive removal, annotated with whether each timing was taken under competing CPU load. (continued)

```

sampled_cpu_percent_max : 100
sampled_n_python_procs_max : 49
  sampled_job_tags : B2
    n_load_samples : 76
      contended : True
cost_analysis_eligible : False
  remeasured_clean : False
shap_seconds_original_contended :
  f1 :
    mcc :
      removed_feature :
        top_feature :
          hyperparameter_regime :

```

[record 15 of 24]

```

  dataset : PhiUSIIL
  iteration : 14
  n_features : 36
  accuracy : 0.989191
  fit_seconds : 33
  shap_seconds : 2022.1
  mean_tree_depth : 42.442
  total_nodes : 2247592
  window_start : 16:22:37
  window_end : 16:56:53
  cache_replay : False
  source_log : a13_resume.log
  run_date : 2026-09-11
  window_reconstructed :
  competing_jobs :
  contention_evidence :
  sampled_cpu_percent_mean : 12.92
  sampled_cpu_percent_max : 100
  sampled_n_python_procs_max : 25
  sampled_job_tags :
  n_load_samples : 97
  contended : False
  cost_analysis_eligible : True
  remeasured_clean : False
shap_seconds_original_contended :
  f1 :
    mcc :
      removed_feature :
        top_feature :
          hyperparameter_regime :

```

[record 16 of 24]

```

  dataset : UCI Website Phishing
  iteration : 0
  n_features : 9
  accuracy : 0.906404
  fit_seconds : 1.77
  shap_seconds : 0.01
  mean_tree_depth : 11.6933
  total_nodes : 54348
  window_start :
  window_end :
  cache_replay : False
  source_log : screening
  run_date :
  window_reconstructed :
  competing_jobs :

```

Table T12. Explanation cost profile for progressive removal, annotated with whether each timing was taken under competing CPU load. (continued)

```

    contention_evidence : not_assessed
    sampled_cpu_percent_mean :
    sampled_cpu_percent_max :
    sampled_n_python_procs_max :
    sampled_job_tags :
    n_load_samples :
    contended :
    cost_analysis_eligible :
    remeasured_clean :
shap_seconds_original_contended :
    f1 : 0.924303
    mcc : 0.805681
    removed_feature :
    top_feature : SFH
    hyperparameter_regime : retuned_per_feature_set

```

[record 17 of 24]

```

    dataset : UCI Website Phishing (variant)
    iteration : 0
    n_features : 9
    accuracy : 0.920213
    fit_seconds : 1.23
    shap_seconds : 0.01
    mean_tree_depth : 9.48
    total_nodes : 9690
    window_start :
    window_end :
    cache_replay : False
    source_log : screening
    run_date :
    window_reconstructed :
    competing_jobs :
    contention_evidence : not_assessed
    sampled_cpu_percent_mean :
    sampled_cpu_percent_max :
    sampled_n_python_procs_max :
    sampled_job_tags :
    n_load_samples :
    contended :
    cost_analysis_eligible :
    remeasured_clean :
shap_seconds_original_contended :
    f1 : 0.930233
    mcc : 0.837552
    removed_feature :
    top_feature : SFH
    hyperparameter_regime : retuned_per_feature_set

```

[record 18 of 24]

```

    dataset : Mendeley (Hannousse)
    iteration : 0
    n_features : 87
    accuracy : 0.965598
    fit_seconds : 2.96
    shap_seconds : 0.01
    mean_tree_depth : 21.58
    total_nodes : 472714
    window_start :
    window_end :
    cache_replay : False
    source_log : screening
    run_date :

```

Table T12. Explanation cost profile for progressive removal, annotated with whether each timing was taken under competing CPU load. (continued)

```

window_reconstructed :
  competing_jobs :
    contention_evidence : not_assessed
sampled_cpu_percent_mean :
sampled_cpu_percent_max :
sampled_n_python_procs_max :
  sampled_job_tags :
    n_load_samples :
      contended :
cost_analysis_eligible :
  remeasured_clean :
shap_seconds_original_contended :
  f1 : 0.965876
  mcc : 0.931339
  removed_feature :
    top_feature : page_rank
hyperparameter_regime : retuned_per_feature_set

```

[record 19 of 24]

```

dataset : Mendeley (Tan)
iteration : 0
n_features : 48
  accuracy : 0.987333
fit_seconds : 1.9
shap_seconds : 0.01
mean_tree_depth : 19.9667
total_nodes : 234594
window_start :
window_end :
cache_replay : False
source_log : screening
run_date :
window_reconstructed :
  competing_jobs :
    contention_evidence : not_assessed
sampled_cpu_percent_mean :
sampled_cpu_percent_max :
sampled_n_python_procs_max :
  sampled_job_tags :
    n_load_samples :
      contended :
cost_analysis_eligible :
  remeasured_clean :
shap_seconds_original_contended :
  f1 : 0.987325
  mcc : 0.974668
  removed_feature : PctExtHyperlinks
  top_feature : PctExtHyperlinks
hyperparameter_regime : retuned_per_feature_set

```

[record 20 of 24]

```

dataset : Mendeley (Tan)
iteration : 1
n_features : 47
  accuracy : 0.982667
fit_seconds : 1.85
shap_seconds : 338.12
mean_tree_depth : 20
total_nodes : 271312
window_start :
window_end :
cache_replay : False

```

Table T12. Explanation cost profile for progressive removal, annotated with whether each timing was taken under competing CPU load. (continued)

```

        source_log : screening
        run_date :
    window_reconstructed :
        competing_jobs :
        contention_evidence : not_assessed
    sampled_cpu_percent_mean :
    sampled_cpu_percent_max :
    sampled_n_python_procs_max :
        sampled_job_tags :
        n_load_samples :
        contended :
    cost_analysis_eligible :
        remeasured_clean :
    shap_seconds_original_contended :
        f1 : 0.982667
        mcc : 0.965333
    removed_feature : PctExtNullSelfRedirectHyperlinksRT
    top_feature : PctExtNullSelfRedirectHyperlinksRT
    hyperparameter_regime : tuned_on_full_feature_set

```

[record 21 of 24]

```

        dataset : Mendeley (Tan)
        iteration : 2
        n_features : 46
        accuracy : 0.980667
    fit_seconds : 2.32
    shap_seconds : 356.45
    mean_tree_depth : 20
    total_nodes : 310536
    window_start :
    window_end :
    cache_replay : False
    source_log : screening
    run_date :
    window_reconstructed :
    competing_jobs :
    contention_evidence : not_assessed
    sampled_cpu_percent_mean :
    sampled_cpu_percent_max :
    sampled_n_python_procs_max :
    sampled_job_tags :
    n_load_samples :
    contended :
    cost_analysis_eligible :
    remeasured_clean :
    shap_seconds_original_contended :
        f1 : 0.980654
        mcc : 0.961334
    removed_feature : PctNullSelfRedirectHyperlinks
    top_feature : PctNullSelfRedirectHyperlinks
    hyperparameter_regime : tuned_on_full_feature_set

```

[record 22 of 24]

```

        dataset : Mendeley (Tan)
        iteration : 3
        n_features : 45
        accuracy : 0.975333
    fit_seconds : 2.24
    shap_seconds : 372.43
    mean_tree_depth : 20
    total_nodes : 343486
    window_start :

```

Table T12. Explanation cost profile for progressive removal, annotated with whether each timing was taken under competing CPU load. (continued)

```

        window_end :
        cache_replay : False
        source_log : screening
        run_date :
    window_reconstructed :
        competing_jobs :
        contention_evidence : not_assessed
    sampled_cpu_percent_mean :
    sampled_cpu_percent_max :
    sampled_n_python_procs_max :
        sampled_job_tags :
        n_load_samples :
        contended :
    cost_analysis_eligible :
        remeasured_clean :
    shap_seconds_original_contended :
        f1 : 0.975284
        mcc : 0.950674
    removed_feature : FrequentDomainNameMismatch
    top_feature : FrequentDomainNameMismatch
    hyperparameter_regime : tuned_on_full_feature_set

```

[record 23 of 24]

```

        dataset : Mendeley (Tan)
        iteration : 4
        n_features : 44
        accuracy : 0.972
        fit_seconds : 2.22
        shap_seconds : 381.82
    mean_tree_depth : 20
    total_nodes : 382488
    window_start :
    window_end :
    cache_replay : False
    source_log : screening
    run_date :
    window_reconstructed :
        competing_jobs :
        contention_evidence : not_assessed
    sampled_cpu_percent_mean :
    sampled_cpu_percent_max :
    sampled_n_python_procs_max :
        sampled_job_tags :
        n_load_samples :
        contended :
    cost_analysis_eligible :
        remeasured_clean :
    shap_seconds_original_contended :
        f1 : 0.972037
        mcc : 0.944003
    removed_feature : PctExtResourceUrls
    top_feature : PctExtResourceUrls
    hyperparameter_regime : tuned_on_full_feature_set

```

[record 24 of 24]

```

        dataset : Mendeley (Tan)
        iteration : 5
        n_features : 43
        accuracy : 0.964667
        fit_seconds : 2.19
        shap_seconds : 414.92
    mean_tree_depth : 20

```

Table T12. Explanation cost profile for progressive removal, annotated with whether each timing was taken under competing CPU load. (continued)

```
total_nodes : 423066
window_start :
window_end :
cache_replay : False
source_log : screening
run_date :
window_reconstructed :
competing_jobs :
contention_evidence : not_assessed
sampled_cpu_percent_mean :
sampled_cpu_percent_max :
sampled_n_python_procs_max :
sampled_job_tags :
n_load_samples :
contended :
cost_analysis_eligible :
remeasured_clean :
shap_seconds_original_contended :
f1 : 0.965017
mcc : 0.929519
removed_feature :
top_feature : ExtMetaScriptLinkRT
hyperparameter_regime : tuned_on_full_feature_set
```

Table T13. Directional agreement per shared concept, ranked by how much the disagreement matters.

concept	rho_phiusiil	rho_uci	direction_agrees	abs_rho_phiusiil	abs_rho_uci	status	importance_weight	headline
anchor_external_ratio	0.0872799	-0.701207	False	0.0872799	0.701207	INVERTED	0.701207	True
iframe	0.521332	0.00339352	False	0.521332	0.00339352	NO_SIGNAL_UCI	0.521332	True
ssl_state	-0.612415	-0.735814	True	0.612415	0.735814	agrees	0.735814	False
url_length	-0.353994	-0.0533316	True	0.353994	0.0533316	agrees	0.353994	False
subdomain_depth	-0.114149	-0.304692	True	0.114149	0.304692	agrees	0.304692	False
form_handler	0.167309	-0.21919	False	0.167309	0.21919	INVERTED	0.21919	False
popup	0.192546	-8.58868e-05	False	0.192546	8.58868e-05	NO_SIGNAL_UCI	0.192546	False
ip_in_domain	-0.060549	-0.0941601	True	0.060549	0.0941601	agrees	0.0941601	False
url_redirect	0.0480144	0.0201135	True	0.0480144	0.0201135	agrees	0.0480144	False

Table T14. Concepts retained and lost when PhiUSIL is restricted to its leak-controlled feature set.

concept	status	reason	rho_uci	rho_phiusiil	polarity_status
ip_in_domain	retained_after_deleaking		-0.0941601	-0.060549	agrees
url_length	retained_after_deleaking		-0.0533316	-0.353994	agrees
subdomain_depth	retained_after_deleaking		-0.304692	-0.114149	agrees
iframe	retained_after_deleaking		0.00339352	0.521332	NO_SIGNAL_UCI
popup	retained_after_deleaking		-8.58868e-05	0.192546	NO_SIGNAL_UCI
url_redirect	retained_after_deleaking		0.0201135	0.0480144	agrees
form_handler	retained_after_deleaking		-0.21919	0.167309	INVERTED
ssl_state	LOST_to_deleaking	source feature(s) removed by de-leaking: ['IsHTTPS']	-0.735814	-0.612415	agrees
anchor_external_ratio	LOST_to_deleaking	source feature(s) removed by de-leaking: ['NoOfExternalRef', 'NoOfSelfRef']	-0.701207	0.0872799	INVERTED

Table T16. Runtime observations for the reproducibility section, each labelled clean or contended.

```
[record 1 of 1]
  observation : uci_full E2 explanation cost rose ~3.5x against the same data under the previous
                hyperparameters
measured_before_seconds : 447
measured_after_seconds : 1581.3
  ratio : 3.54
  driver : RandomForest n_estimators 100 -> 500
  driver_exact : True
  driver_detail : uci_full received its own hyperparameter search in A4 (justified: it carries 11,055 rows
                  against uci_dedup's 5,849). That search selected 500 trees where the earlier uci parameters
                  used 100 - a 5x increase in the forest the explainer must walk.
  mechanism : Interventional TreeSHAP cost scales with the total node count of the model being explained,
              so explanation cost tracks model complexity regardless of WHY the complexity changed.
  significance : An INDEPENDENT confirmation of the B5b mechanism through a different driver. In A1.3 node
                 count rose because features were removed and the boundary hardened; here it rose because a
                 hyperparameter search chose a larger forest. Same relationship, unrelated cause - which is
                 what makes explanation cost a general proxy for model complexity rather than an artifact of
                 the removal procedure.
before_timing_clean : False
after_timing_clean : False
  contention_note : NEITHER timing is uncontended: the 447s baseline ran alongside A1.3, and the 1581s figure
                  ran alongside table-building commands (load sampler shows job tags A4A5+B2 across
                  18:38-19:05). The 5x tree count is exact and structural; the ~3.5x ratio is approximate and
                  should be quoted as 'roughly 3.5x' or re-measured on an idle machine before being stated
                  precisely.
  n_explained : 5527
    source : A5 uci_full seed 42/43, half A
```